

Name: _____

Date: _____

FRACTIONS ALL OPERATIONS, (FRACTIONS OF AMOUNTS)

LO: I can find a fraction of a given amount by dividing and multiplying.

Finding a fraction of an amount

To find a fraction of an amount, divide the amount by the denominator then multiply by the numerator.

Method: $\frac{a}{b}$ of $n = n \div b \times a$. Always divide first to keep numbers small.

Quick examples

●	$\frac{1}{4}$ of 60 = 15	$60 \div 4 = 15$
●	$\frac{2}{5}$ of 30 = 12	$30 \div 5 = 6$, then 6×2
●	$\frac{3}{4}$ of 80 = 60	$80 \div 4 = 20$, then 20×3
●	$\frac{2}{5}$ of 30 = 6	forgot to multiply by numerator
●	$\frac{3}{4}$ of 80 = 240	multiplied by both 3 and 4

Key idea

Divide by the denominator first to find one part, then multiply by the numerator to find the required number of parts.

$$\frac{a}{b} \text{ of } n = n \div b \times a$$

Divide by bottom, multiply by top

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - unit fraction

Find $\frac{1}{5}$ of 45

Divide by the denominator: $45 \div 5$
= 9

$$\frac{1}{5} \text{ of } 45 = 9$$

$$= 9$$

For unit fractions, just divide by the denominator.

Example 2 - non-unit fraction

Find $\frac{3}{4}$ of 60

Divide by 4: $60 \div 4 = 15$
Multiply by 3: $15 \times 3 = 45$

$$\frac{3}{4} \text{ of } 60 = 45$$

$$= 45$$

Divide by the bottom number, multiply by the top number.

Example 3 - fraction of money

Find $\frac{2}{3}$ of £18

Divide by 3: $18 \div 3 = 6$
Multiply by 2: $6 \times 2 = 12$

$$\frac{2}{3} \text{ of } £18 = £12$$

$$= £12$$

Same method works for money and measurements.

Example 4 - finding the whole

$\frac{3}{5}$ of a number is 24. Find the number.

If $\frac{3}{5} = 24$, find $\frac{1}{5}$: $24 \div 3 = 8$

Find the whole: $8 \times 5 = 40$

The number is 40

$$= 40$$

Divide by the numerator to find one part, then multiply by the denominator.

YOUR TURN – PRACTICE (deliberate practice)

1. Unit fractions of amounts

Find each fraction of the amount.

- a) $\frac{1}{3}$ of 27
- b) $\frac{1}{4}$ of 48
- c) $\frac{1}{6}$ of 42
- d) $\frac{1}{8}$ of 56

2. Non-unit fractions

Find the fraction of each amount.

- a) $\frac{2}{5}$ of 35
- b) $\frac{3}{4}$ of 44
- c) $\frac{4}{7}$ of 49
- d) $\frac{5}{8}$ of 72

3. Fractions of money

Find the fraction of each amount.

- a) $\frac{3}{5}$ of £40
- b) $\frac{2}{3}$ of £63
- c) $\frac{5}{6}$ of £54
- d) $\frac{7}{10}$ of £120

4. Finding the whole amount

Work backwards to find the original amount.

- a) $\frac{1}{4}$ of a number = 9. Find the number.
- b) $\frac{2}{5}$ of a number = 18. Find the number.
- c) $\frac{3}{7}$ of a number = 21. Find the number.
- d) $\frac{5}{8}$ of a number = 30. Find the number.

5. Mixed problems

Solve each problem.

- a) $\frac{3}{10}$ of 80
- b) $\frac{4}{9}$ of 63
- c) $\frac{2}{3}$ of £99
- d) $\frac{5}{6}$ of a number = 35. Find it.
- e) $\frac{7}{8}$ of 64

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $\frac{2}{5}$ of 30 means $30 \div 5 \times 2 = 12$

☐

Correct answer: _____

Reason: _____

b) $\frac{2}{5}$ of 30 means $30 \div 5 = 6$ ☐

Correct answer: _____

Reason: _____

c) $\frac{3}{4}$ of 80 = 60 ☐

Correct answer: _____

Reason: _____

d) Finding a fraction of an amount always gives a smaller number ☐

Correct answer: _____

Reason: _____

e) To find the whole, divide by numerator then multiply by denominator ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A school has **840 pupils**. $\frac{3}{7}$ are in KS3 and $\frac{2}{5}$ of the remainder are in KS4. How many pupils are in KS4? How many are in KS5?

Expression: _____

Simplified: _____

b) Jake has **£120**. He spends $\frac{1}{4}$ on a book, then $\frac{2}{3}$ of what remains on a game. How much does he have left?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

